Experiment-12

Write a python program for Tic Tac Toe game

Aim

To write and execute a Python program to implement the Tic Tac Toe game for two players (X and O) on a 3 x 3 board.

Algorithm

1. Create a 3 x 3 board with empty cells numbered from 1 to 9.
2. Let Player X start the game first.
3. Display the current board.
4. Ask the current player to enter a position between 1 and 9.
5. Check whether the position is valid and empty, otherwise ask again.
6. Place the player's symbol (X or O) on the board.
7. Check the winning conditions:
    - Three same symbols in any row.
    - Three same symbols in any column.
    - Three same symbols in any diagonal.
8. If a player wins, display the board and the winner, and stop the game.
9. Otherwise, switch to the other player and repeat the steps.
10. If all 9 cells are filled and no one has won, display that the game is a draw.

Coding

ex-12.py - D:/ex-12.py (3.10.0)

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```
# Tic Tac Toe Game (Two Players)

board = [" "] * 9

def show():
    print()
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])

def winner(p):
    lines = [(0, 1, 2), (3, 4, 5), (6, 7, 8),
             (0, 3, 6), (1, 4, 7), (2, 5, 8),
             (0, 4, 8), (2, 4, 6)]
    for a, b, c in lines:
        if board[a] == board[b] == board[c] == p:
            return True
    return False

print("TIC TAC TOE GAME")
print("Positions are numbered 1 to 9")
player = "X"

for turn in range(9):
    show()
    move = int(input("Player " + player + " enter position : ")) - 1

    # Check that the move is valid
    while move < 0 or move > 8 or board[move] != " ":
        print("Invalid move, try again")
        move = int(input("Player " + player + " enter position : ")) - 1
```

ex-12.py - D:/ex-12.py (3.10.0)

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```
    board[move] = player

    if winner(player):
        show()
        print()
        print("Player", player, "wins!")
        break

    if player == "X":
        player = "O"
    else:
        player = "X"
else:
    print("Game is a Draw")
```

Output


IDLE Shell 3.10.0
File Edit Shell Debug Options Window Help
Python 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: D:/ex-12.py ==========
TIC TAC TOE GAME
Positions are numbered 1 to 9
| |
--+---+--
| |
--+---+--
| |
Player X enter position : 5
| |
--+---+--
| X |
--+---+--
| |
Player O enter position : 1
O | |
--+---+--
| X |
--+---+--
| |
Player X enter position : 7


IDLE Shell 3.10.0
File Edit Shell Debug Options Window Help
O | |
--+---+--
| X |
--+---+--
X | |
Player O enter position : 5
Invalid move, try again
Player O enter position : 9
O | |
--+---+--
| X |
--+---+--
X | | O
Player X enter position : 3
O | | X
--+---+--
| X |
--+---+--
X | | O
Player X wins!
>>>

Result

The Tic Tac Toe game was successfully implemented in Python. In the sample run, Player X won the game by occupying the diagonal positions 3, 5 and 7.